

As noted, the value of the bond ladder does fluctuate over time with changing interest rates, but this is immaterial to the success of the plan if the bonds are held to maturity. At maturity they pay the full expected value, and earlier unrealized gains or losses do not materialize. Investors financing a retirement goal can happily ignore the fluctuating value of their individual bonds, knowing that the desired cash flows will be provided from bond coupon payments and the return of principal at maturity. When laddered bonds are held to maturity, cash flows are known, and there is no realized interest-rate risk. In fact, rising interest rates could even help with issues such as reducing the overall IRS required minimum distribution (RMD) amounts for bonds in tax-deferred accounts.

With a bond fund, the concept of duration implies an investor is made whole after a rate increase once a time period matching the fund's duration has passed, because that investor is able to reinvest coupon payments at higher interest rates to offset the capital losses on bonds. However, that conclusion assumes the portfolio is not funding a spending goal. If those coupons are being used for spending and if some shares of the bond fund are sold to cover spending, then the bond portfolio will not be able to recover through its ability to reinvest cash flows at a new higher interest rate. Portfolio returns would need to be even higher to offset the loss in ability to fully reinvest funds at higher rates. This makes duration matching for household investors using constant-duration bond funds much more difficult in practice. Only a few commercial solutions exist.

Time segmentation does provide a practical way for retirees to duration-match their spending goals. But the idea that time segmentation reduces sequence risk is less clear-cut. Time segmentation is a probability-based approach in that its advocates are confident that sufficient upside growth will take place before growth assets need to be sold to support spending. The safety-first side reminds us that there is no guarantee this will actually happen. Growth assets may still have to be sold at a loss, depending on the mechanics for when it becomes necessary to rebalance into bonds. The bond ladder may be depleted before the growth portfolio has recovered sufficiently. Sequence risk may still materialize if assets are sold at a loss. We need to empirically investigate the importance of this matter, which we will now do.